

WarpX Run (16min 20s) with scaled inputs from Myers 2021 <https://www.sciencedirect.com/science/article/pii/S0167819121000818#sec4>

Parameter	Value	Where set
Grid	256 × 256 × 256 cells	<code>my_constants.nx = 256</code>
Timesteps	500	<code>my_constants.nt = 500</code>
Domain	40 μm cube (−20 to +20 μm each axis)	<code>geometry.prob_lo/hi</code>
Species	2 (electrons + protons)	<code>particles.species_names</code>
Particles per cell	8 per species (2×2×2) → 16 total	<code>num_particles_per_cell_each_dim</code>
Total particles	~537 M (256 ³ × 16)	derived
Density	10 ²⁵ m ^{−3} each species	<code>.density</code>
Thermal velocity	0.5 (dimensionless β _y)	<code>my_constants.uth = 0.5</code>
Shape order	3 (cubic B-spline)	<code>algo.particle_shape = 3</code>
Current deposition	Esirkepov	<code>algo.current_deposition</code>
Pusher	Boris	<code>algo.particle_pusher</code>
Maxwell solver	Yee (FDTD)	<code>algo.maxwell_solver</code>
Boundaries	Periodic (fields + particles, all axes)	<code>boundary.*</code>
I/O	None	<code>diagnostics.diags_names</code> absent

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+ 92.86% 0.01% warpx.3d.NOMPI. warpx.3d.NOMPI.OMP.DP.PDP.OPMD.EB.QED [.] PhysicalParticleContainer::Evolve(abiasttr::fields::MultiFabRegister&◆
+ 51.59% 42.69% warpx.3d.NOMPI. warpx.3d.NOMPI.OMP.DP.PDP.OPMD.EB.QED [.] PhysicalParticleContainer::PushPX(WarpXParIter&, amrex::FArrayBox co
+ 41.25% 0.00% warpx.3d.NOMPI. warpx.3d.NOMPI.OMP.DP.PDP.OPMD.EB.QED [.] WarpXParticleContainer::DepositCurrent(WarpXParIter&, amrex::PODVect
+ 40.81% 40.79% warpx.3d.NOMPI. warpx.3d.NOMPI.OMP.DP.PDP.OPMD.EB.QED [.] void doEsirkepovDepositionShapeN<3>(GetParticlePosition<PIdx> const&
+ 8.74% 8.73% warpx.3d.NOMPI. warpx.3d.NOMPI.OMP.DP.PDP.OPMD.EB.QED [.] UpdateMomentumBoris(double&, double&, double&, double, double, doub
+ 6.88% 0.00% warpx.3d.NOMPI. libgomp.so.1.0.0 [.] 0x000040001b002faf
+ 6.59% 0.00% warpx.3d.NOMPI. libc.so.6 [.] thread_start
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+ 3.86% 1.33% warpx.3d.NOMPI. warpx.3d.NOMPI.OMP.DP.PDP.OPMD.EB.QED [.] amrex::ParticleContainer_impl<amrex::SoAParticle<7, 0>, 7, 0, amrex:
+ 0.95% 0.65% warpx.3d.NOMPI. warpx.3d.NOMPI.OMP.DP.PDP.OPMD.EB.QED [.] bool amrex::ParticleContainer_impl<amrex::SoAParticle<7, 0>, 7, 0, a
+ 0.92% 0.85% warpx.3d.NOMPI. warpx.3d.NOMPI.OMP.DP.PDP.OPMD.EB.QED [.] amrex::IntVectND<3> amrex::ParticleContainer_impl<amrex::SoAParticle
+ 0.52% 0.52% warpx.3d.NOMPI. warpx.3d.NOMPI.OMP.DP.PDP.OPMD.EB.QED [.] amrex::BoxND<3>::contains(amrex::IntVectND<3> const&) const
0.50% 0.50% warpx.3d.NOMPI. warpx.3d.NOMPI.OMP.DP.PDP.OPMD.EB.QED [.] void FiniteDifferenceSolver::EvolveBCartesian<CartesianYeeAlgorithm>
0.38% 0.00% warpx.3d.NOMPI. warpx.3d.NOMPI.OMP.DP.PDP.OPMD.EB.QED [.] main
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92.86% 0.01% PhysicalParticleContainer::Evolve
51.59% 42.69% PhysicalParticleContainer::PushPX wrapper for gather + push
41.25% 0.00% WarpXParticleContainer::DepositCurrent FIELD GATHER KERNEL
40.81% 40.79% doEsirkepovDepositionShapeN<3> ESIRKEPOV Deposition KERNEL
8.74% 8.73% UpdateMomentumBoris actual Boris KERNEL small and fast
0.50% 0.50% FiniteDifferenceSolver::EvolveBCartesian<CartesianYeeAlgorithm>
```

Kernel	Self%	Physics phase
PushPX	42.69%	Boris push wrapper — but the UpdateMomentumBoris line below shows only 8.73% of that is the actual Boris kernel. The remaining ~34% is field gather (inlined into PushPX)
doEsirkepovDepositionShapeN<3>	40.79%	Current deposition
UpdateMomentumBoris	8.73%	Boris momentum update
AMReX particle infrastructure	~2.5% (1.33 + 0.65 + 0.85 + ...)	Iterator/bookkeeping
EvolveBCartesian	0.50%	Maxwell B-field solve